

4(a)	cohesion ; adhesion ;	2
4(b)(i)	any two from: 1 higher rate of transpiration during the day / lower rate of transpiration during the night / AW ; A flow of xylem sap <i>for transpiration</i> 2 stomata are open during the day ; ora <i>must be in context of transpiration</i> <i>if mp1 and mp2 not gained, allow one mark for idea that results are related to pattern of, light / daylight / day, and, dark / night (use an ordinary tick)</i> 3 <i>idea that</i> all three vines were kept in the same conditions so a changed condition that affects transpiration will affect all three ; A all three vines have the same external factors acting that affect transpiration 4 AVP ; e.g. (during daylight) stomata open, to obtain carbon dioxide / for photosynthesis <i>idea that</i>, increasing / decreasing, rate is related to degree of opening of stomata varying with light intensity	2

Question	Answer	Marks
4(b)(ii)	<i>allow (xylem sap) flow rate for rate of transpiration any two from:</i> 1 the greater the (total) area of leaf, the higher the rate of transpiration ; ora A grapevine 1 has largest leaf area and highest rate of transpiration <i>accept other comparisons</i> R ref. to SA:V but allow ecf in mp2 2 the greater the (total) area of leaf, the more stomata are present ; AW A more leaves means more stomata 3 detail of any one grapevine ; e.g. grapevine 1 = highest transpiration rate and, highest number of / most, stomata grapevine 3 = lowest transpiration rate and, lowest number of / least, stomata <i>allow use of comparative data to support mp</i> <i>(if detail includes ref. to (total) leaf area then check to see if mp2 can also be awarded)</i> 4 <i>ref. to</i> relationship between (total) internal surface area and (total) leaf area ; e.g. larger leaf surface area means internal surface area increased 5 <i>ref. to</i> relationship between internal surface areas and rates of evaporation ; 6 AVP ; higher rate of flow of xylem sap for largest leaf area as (overall) more water used in metabolism AW ora (greater area so) more leaves so more xylem vessels ora	2
4(b)(iii)	overcast / cloudy / rain / shade / AW ; A lower wind speed / temperature A higher humidity R if described as stomata close (transpiration is still occurring)	1

Question	Answer	Marks
4(b)(iv)	<i>valid method e.g.</i> draw around / place, leaf on, squared paper / graph paper / grids ; A square ruler count number of (full) squares ; R multiply by two / do it on other side method for part squares ; e.g. add up all part squares and divide by 2 match a larger part square with a smaller and count as one square use graph paper with larger and smaller squares or photocopy leaves, cut out and weigh (cut out) ; divide mass by mass of 1 cm³ of same paper ; or <i>ref. to</i> using an App ; further detail ; e.g. place leaf, flat / on white surface ; take photo of single leaf with smart phone / AW	2

Question	Answer	Marks
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